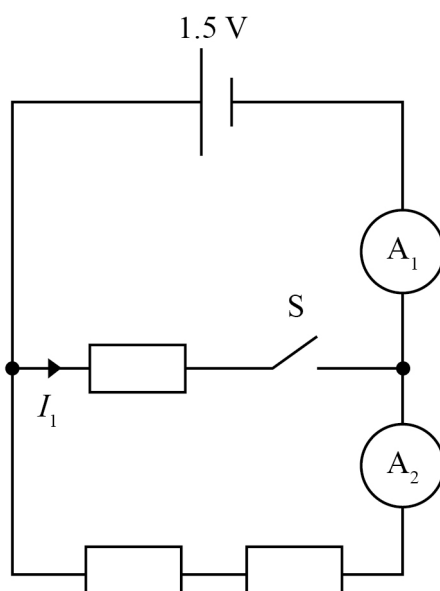


0 4

Figure 2 shows three identical resistors, each with a resistance R , in a circuit connected to a cell with emf 1.5 V

The cell has negligible internal resistance.

Figure 2



0 4 . 1

When switch S is closed the reading of ammeter A_1 is 1.2 A and the reading of ammeter A_2 is 0.40 A

State the value of the current I_1 when switch S is closed.

[1 mark]

$I_1 =$ _____ A

0 4 . 2

Calculate R .

[1 mark]

$R =$ _____ Ω



0 4 . 3

Switch S is returned to the open position.

State and explain the effect this has on the readings of A_1 and A_2 .

[3 marks]

5

Turn over for the next question

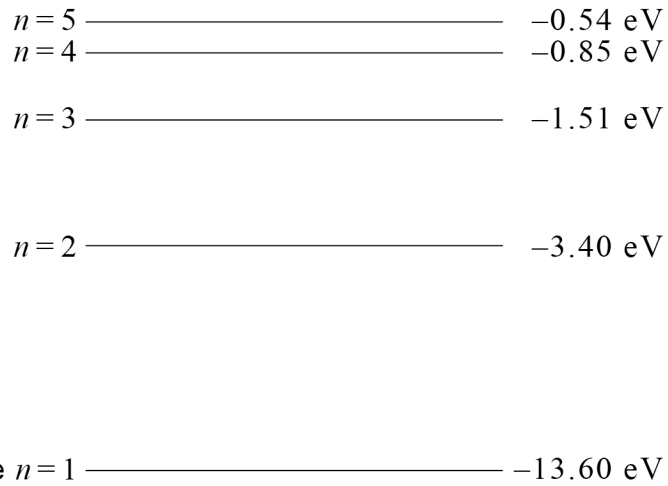
Turn over ►



Don't write

Figure 3 shows some of the energy levels for a hydrogen atom.

Figure 3



A hydrogen atom is in the ground state. It absorbs all the energy of a photon and becomes excited to the $n = 3$ energy level.

Calculate the frequency of the photon absorbed by the hydrogen atom.

[3 marks]

frequency = _____ Hz

State, in eV, the ionisation energy of a hydrogen atom.

[1 mark]

ionisation energy = _____ eV



0 5 . 3

A photon with an energy of 18.4 eV interacts with a hydrogen atom in its ground state.

Describe how the principle of energy conservation could apply in this situation.

[3 marks]

7

Turn over for the next question

Turn over ►



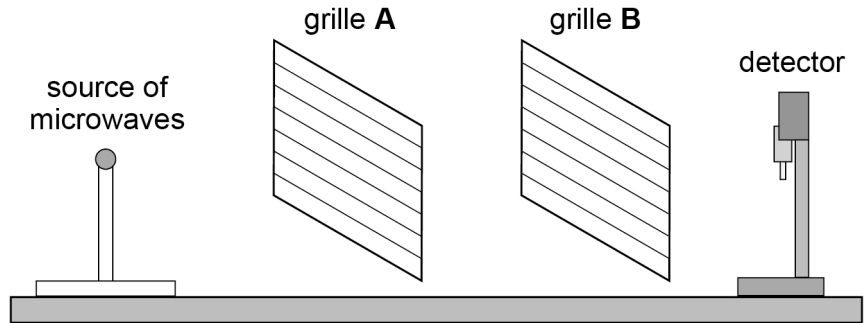
0 6 . 1 Describe the nature of an electromagnetic wave.

[3 marks]



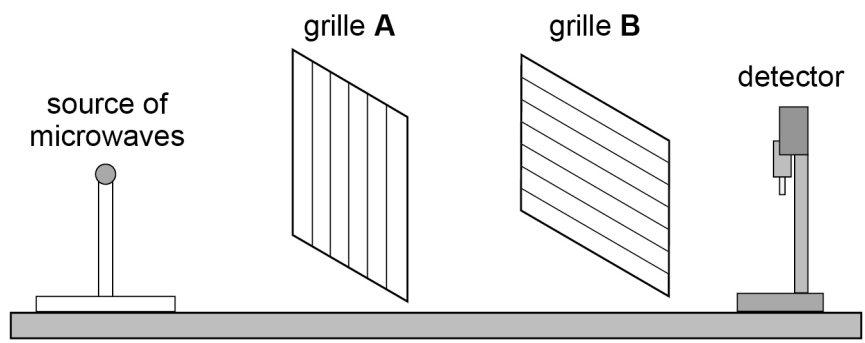
A source of microwaves and a detector are set up with two metal grilles **A** and **B** between them, as shown in **Figure 4**. The reading on a voltmeter connected to the detector is 5 mV. Each grille consists of a series of parallel, thin, metal rods.

Figure 4



Grille **A** is rotated through 90° about a horizontal axis as shown in **Figure 5**.

Figure 5

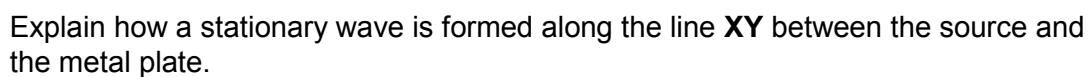


Explain the changes in the voltmeter reading as grille **A** is rotated.

[2 marks]



Figure 6

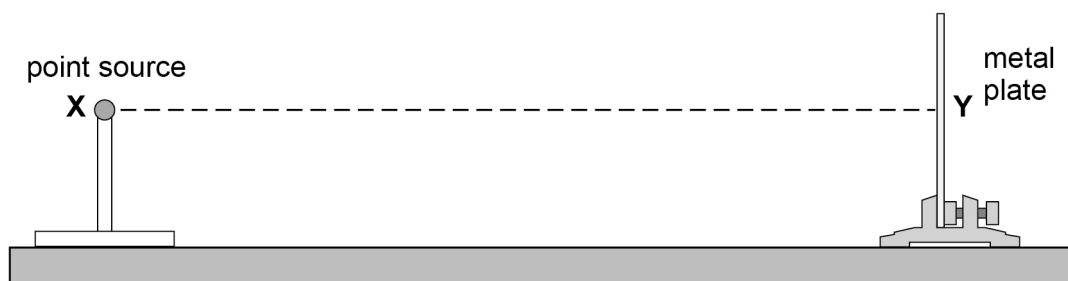


[3 marks]

[illegible]

Sketch on **Figure 7** the shape of the stationary wave formed.

Figure 7



0	6	.	5
---	---	---	---

Explain how the voltmeter readings vary as the detector is moved along the line **XY**.

[2 marks]

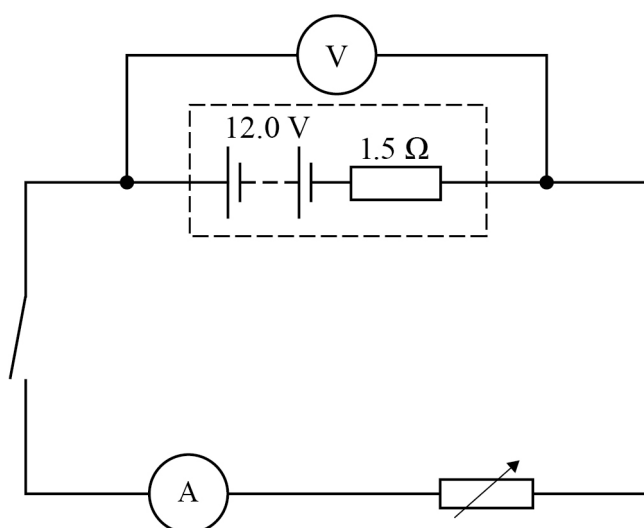
[illegible]

0 7 . 1 Define the electromotive force (emf) of a battery.

[2 marks]

0 7 . 2 A battery with an emf of 12.0 V and an internal resistance of $1.5\ \Omega$ is connected in the circuit shown in **Figure 8**.

Figure 8



The voltmeter reading is 9.0 V when the current in the circuit is 2.0 A

Calculate the resistance of the variable resistor when the voltmeter reads 9.0 V

[2 marks]

resistance = _____ Ω



0 7 . 3 Determine the maximum current that can be provided by the battery.

[2 marks]

maximum current = _____ A

0 7 . 4 With the switch closed the variable resistor is adjusted to obtain a range of ammeter and voltmeter readings. Finally the switch is opened and a final ammeter and voltmeter reading are obtained.

Sketch on **Figure 9** a graph to show the variation of voltmeter reading V with current I .

Label your axes with suitable numerical values.

[2 marks]

Figure 9



Question 7 continues on the next page

0 7 . 5

The battery, when fully charged, can deliver a total charge of $1.15 \times 10^4 \text{ C}$

For a particular application, the fully-charged battery is required to supply a constant current to an external circuit of resistance 0.1Ω for 30 minutes.

Discuss the suitability of the battery for this application.

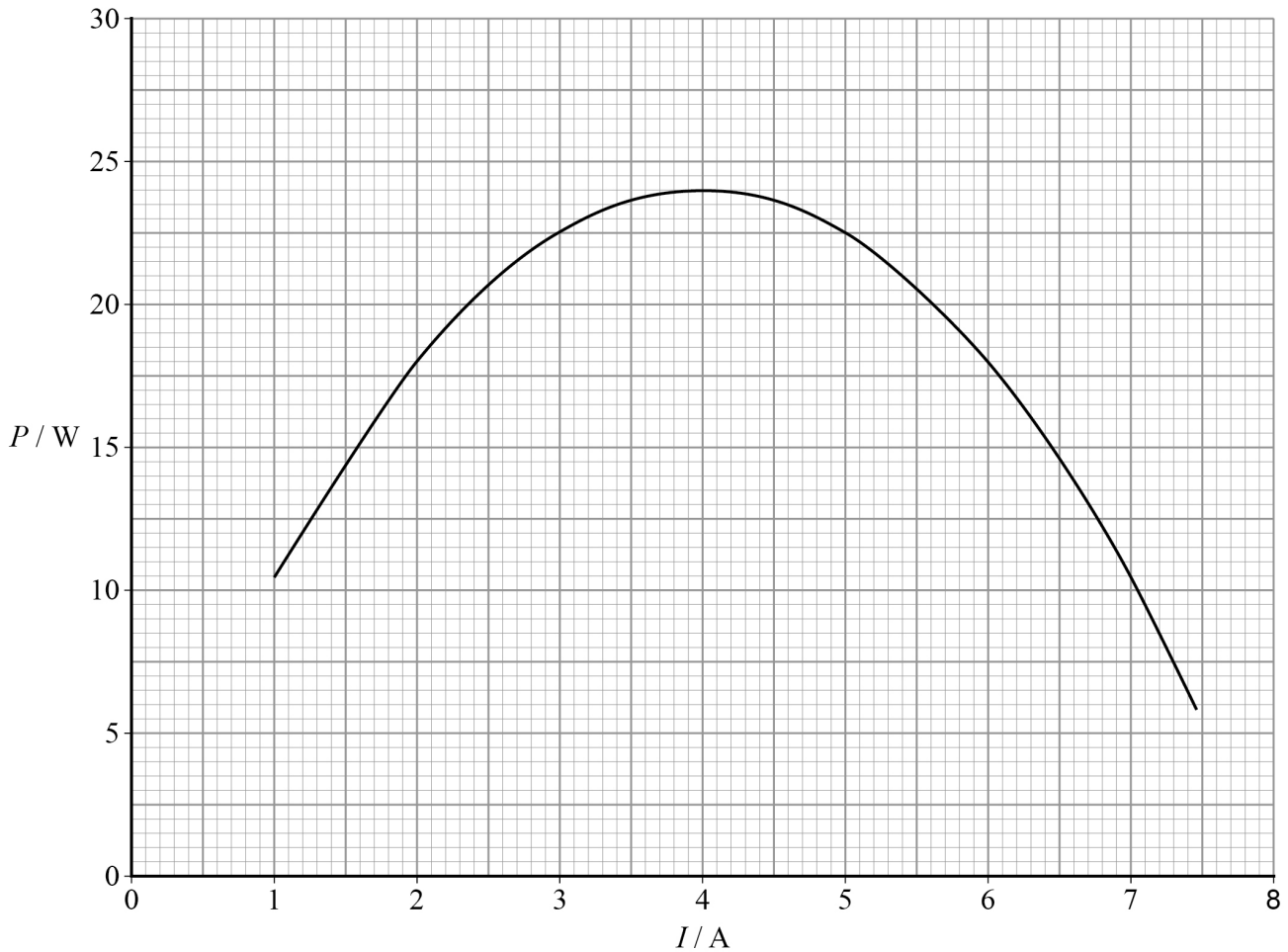
You should use calculations to support your answer.

[3 marks]



Figure 10 shows the variation with current of the power dissipated in the variable resistor.

Figure 10



Calculate using data from **Figure 10** the value of the variable resistance when P is a maximum.

[3 marks]

variable resistance = _____ Ω

Question 7 continues on the next page

0 7 . 7

Figure 10 suggests that as the current increases past 7.5 A, the power dissipated in the variable resistor eventually reaches zero.

Explain why the circuit behaves in this way.

[2 marks]

16

END OF SECTION A



Section B

Answer **all** questions in this section.

*Do not write
outside the
box*

1

1

Which is equivalent to the SI unit of resistivity?

[1 mark]

A $\text{A V}^{-1} \text{m}^{-1}$

☐

B $\text{C}^{-1} \text{V s m}$

☐

C V A m

☐

D C V m s^{-1}

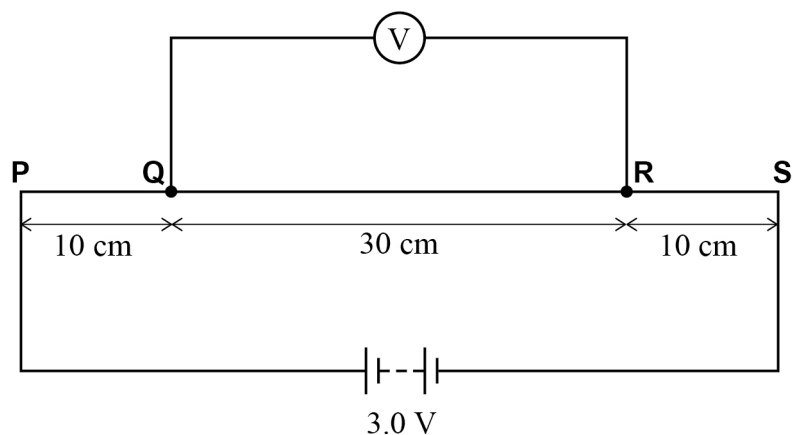
☐

Turn over ►



Don't write

A length **PQRS** of resistance wire of uniform cross-section is connected to a battery of emf 3.0 V and negligible internal resistance.



What is the reading on the voltmeter?

[1 mark]

A 0.6 V

B 1.2 V

C 1.8 V

D 2.4 V

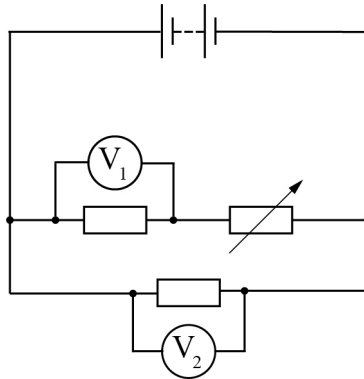
Turn over for the next question

Don't write

Figure 2 shows a circuit containing two fixed resistors, a variable resistor and a battery with negligible internal resistance.

Voltmeters V_1 and V_2 are connected across the fixed resistors.

Figure 2



The resistance of the variable resistor is increased.

State and explain how the readings on V_1 and V_2 are affected by this change.

[4 marks]

V_1 _____

V_2 _____

Don't write

0	4
---	---

A car battery has an emf of 12.2 V and an internal resistance of 0.015 Ω .

0	4	.	1
---	---	---	---

A resistor of resistance $7.6 \times 10^{-3} \Omega$ is connected directly between the terminals of the battery.

Calculate the current in the resistor.

[2 marks]

current = _____ A

0	4	.	2
---	---	---	---

Calculate the rate of heating inside the battery as soon as the resistor is connected between the terminals.

[2 marks]

rate of heating = _____ W

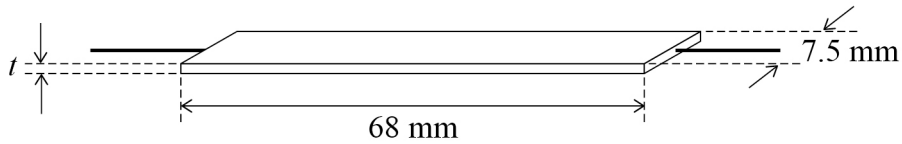
4



Don't write

Figure 10 shows a resistor made from a thin film of material of resistivity $2.8 \times 10^{-3} \Omega \text{ m}$.

Figure 10



The resistor is 68 mm long and 7.5 mm wide.
The resistance of the resistor is 570 Ω .

Calculate, in m, the thickness t of the resistor.

[2 marks]

$t =$ _____ m

Question 10 continues on the next page

The resistor is made into a potential divider that is used in a fuel gauge for a car's fuel tank.

Figure 11A shows the sliding electrical contact made by one end of a pivoted conducting rod. The other end of the rod is attached to a float that is always at the top surface of the fuel in the tank as shown in **Figure 11B** and **Figure 11C**.

A 12 V battery is connected across the resistor. A voltmeter measures the pd across one part of the resistor and acts as the fuel gauge.

Figure 11B shows the fuel gauge in a full fuel tank.

Figure 11C shows the fuel gauge in an almost empty tank.

Figure 11A

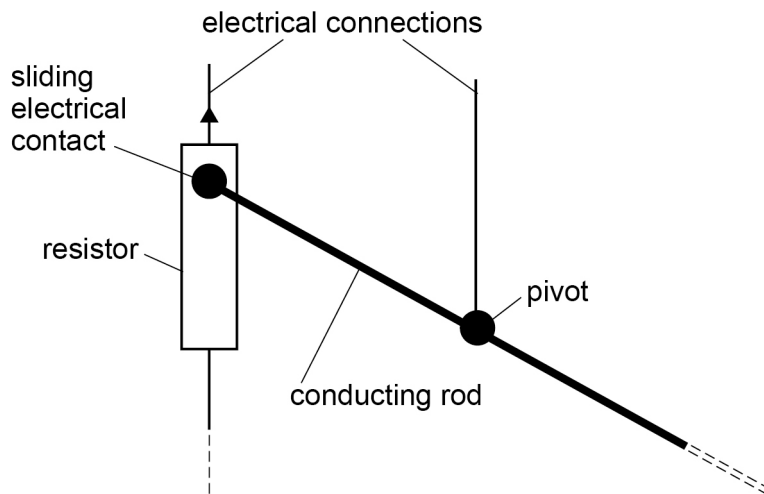


Figure 11B

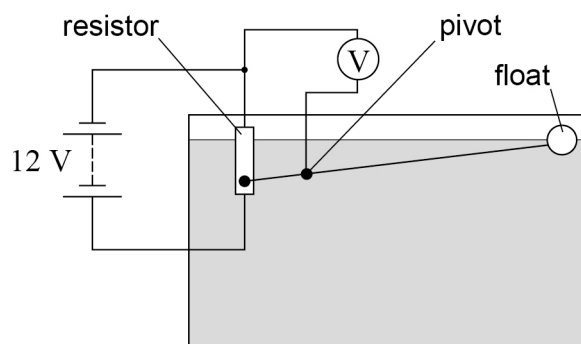
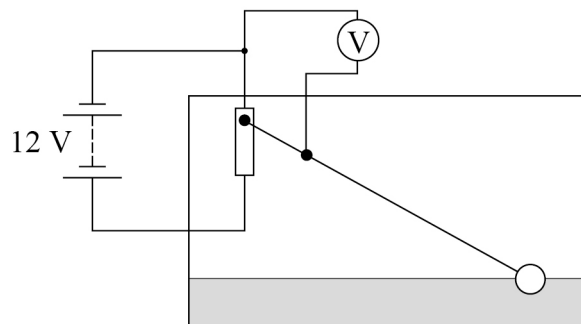


Figure 11C



1 0 . 2 Explain the changes to the voltmeter reading as the level of the fuel in the tank falls.

[2 marks]

1 0 . 3 Calculate the rate of energy transfer in the resistor connected to the 12 V battery.

[2 marks]

rate of energy transfer = _____ W

1 0 . 4 The resistor is changed to one of greater resistance.

State any effect that the change has on the operation of the fuel gauge.

[1 mark]

END OF SECTION B



Section C

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.

For each question, completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown.



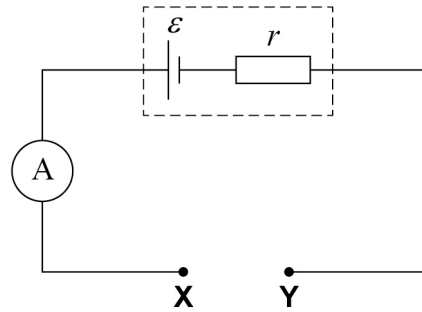
If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown.



You may do your working in the blank space around each question but this will not be marked.
Do **not** use additional pages for this working.

Figure 7 shows a circuit used by a student to determine the emf $\mathcal{E}$ and internal resistance r of a cell.

Figure 7

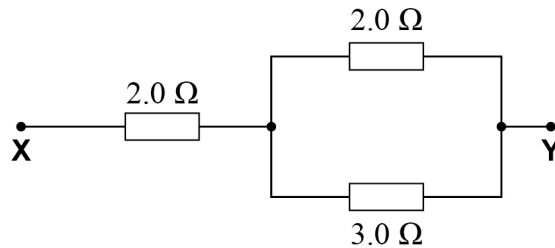


An external resistance R can be connected between **X** and **Y**. Its value can be changed by using different combinations of resistors.

She uses a combination of resistors to make a network with resistances ranging from $1.0\ \Omega$ to $8.0\ \Omega$.

Figure 8 shows one of these networks inserted between **X** and **Y** in the circuit.

Figure 8



Calculate the resistance between **X** and **Y** in **Figure 8**.

[2 marks]

resistance = _____ Ω



The student measures the current I in the ammeter for each value of R and plots a graph of $\frac{1}{I}$ against R .

0 8 . 2 Show that

$$\frac{1}{I} = \frac{R}{\varepsilon} + \frac{r}{\varepsilon}$$

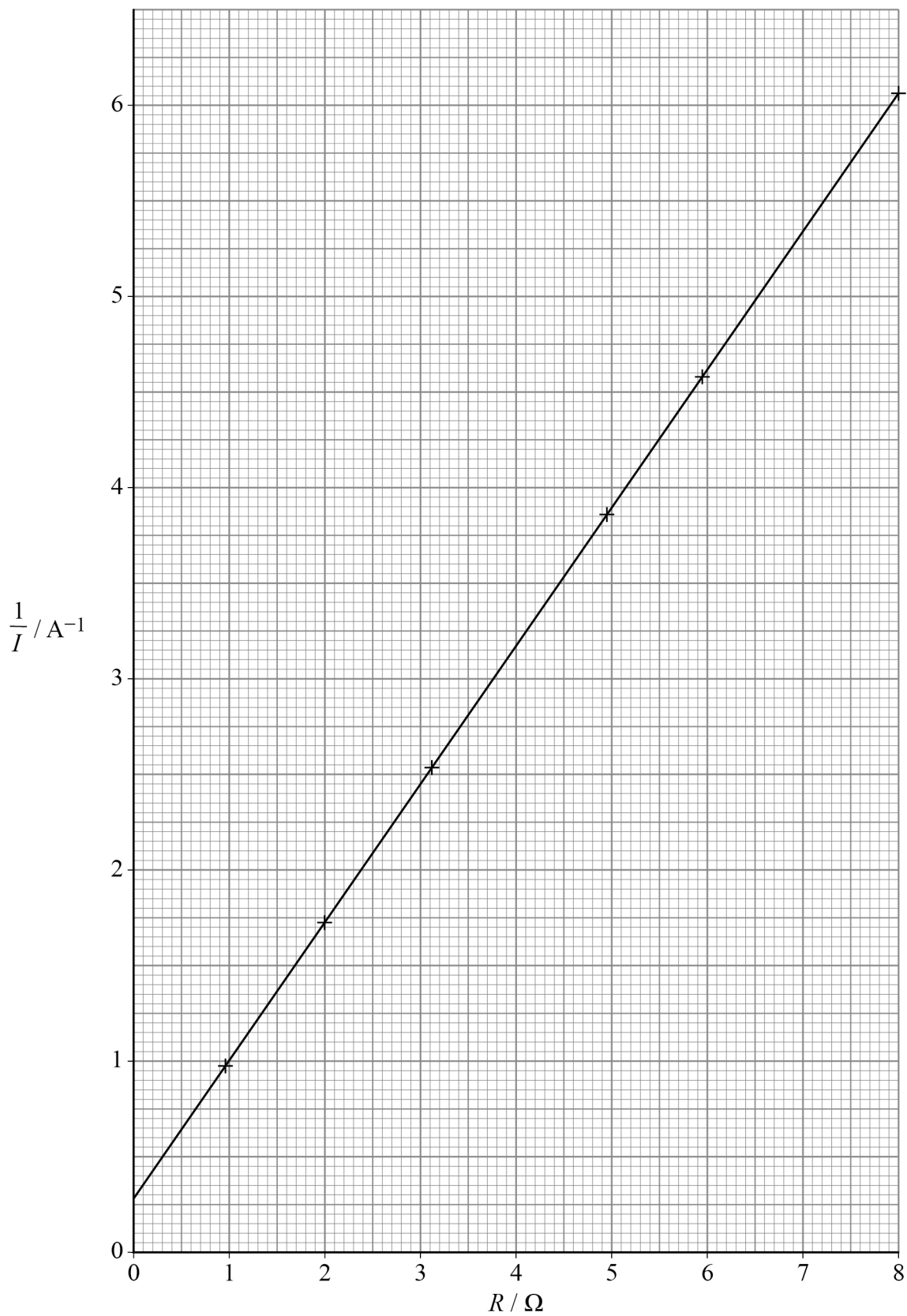
[1 mark]

Question 8 continues on the next page



Figure 9 shows the student's graph.

Figure 9



0	8
---	---

.

3

 Determine ε .

[3 marks]

$\varepsilon =$ _____ V

0	8
---	---

.

4

 Determine r .

[3 marks]

$r =$ _____ Ω

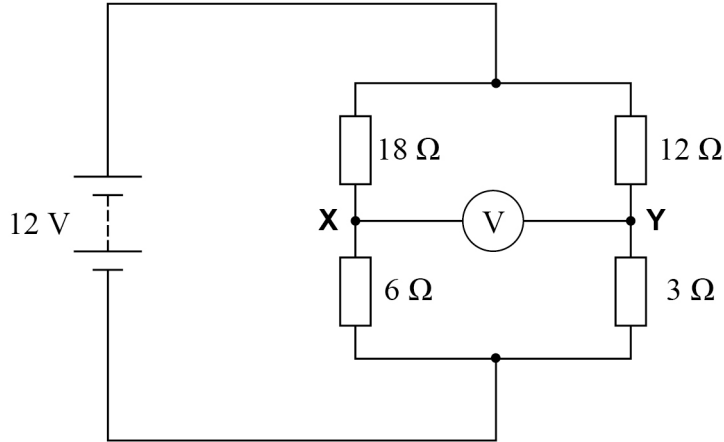
9



Do not write Do not write

1 3

The diagram shows a battery of emf 12 V and negligible internal resistance connected across a network of resistors.



When a voltmeter is connected across **X** and **Y** the reading is

[1 mark]

A 0.6 V .

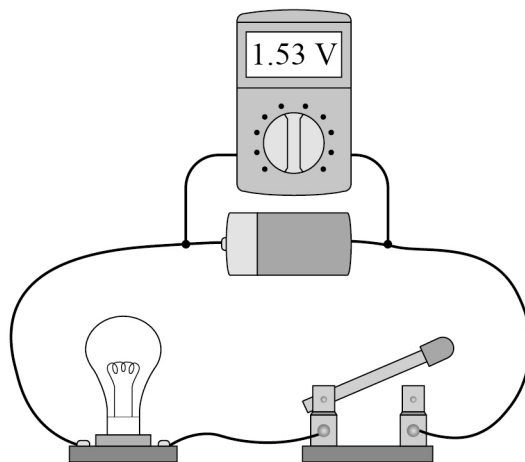
B 6.0 V .

C 9.0 V .

D 9.6 V .

Figure 1 shows an electrical circuit containing a cell, lamp, switch and voltmeter.

Figure 1



The switch is initially open and the voltmeter reading is 1.53 V.
When the switch is closed, the voltmeter reading changes.

Explain this change.

[3 marks]

Do not write

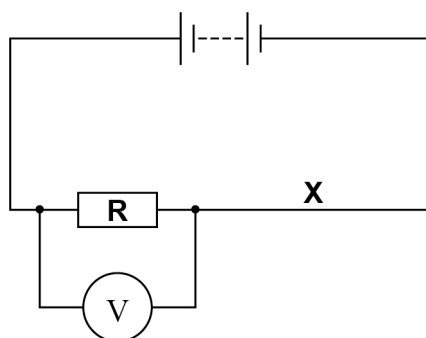
0 5 A student plans to determine the resistance of a fixed resistor **R**.

0 5 . 1 State what is meant by the resistance of **R**.

[1 mark]

The student uses the circuit shown in **Figure 3**.
The internal resistance of the battery is negligible.

Figure 3



The voltmeter reading is V_1 . The voltmeter is ideal.

The student then adds a **non-ideal** ammeter at point **X**.
The voltmeter reading decreases.

The new voltmeter reading is V_2 . The ammeter reading is I .

0 5 . 2 Explain why V_2 is smaller than V_1 .

[3 marks]



0 5 . 3

The student uses V_2 and I to determine the resistance of **R**.

Explain whether this determination is valid.

[1 mark]

5

Turn over for the next question

Turn over ►



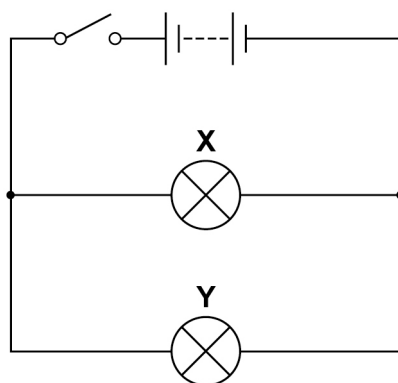
Do not write

0	8
---	---

Figure 11 shows a circuit containing two identical car headlamps **X** and **Y**.

The battery has an emf of 12.8 V and an internal resistance of 61.0 m Ω .

Figure 11



0	8	.	1
---	---	---	---

Both lamps turn on when the switch is closed.
Each lamp now has a resistance of 2.40 Ω .

Show that the total resistance of the circuit is approximately 1.3 Ω .

[2 marks]

0	8	.	2
---	---	---	---

Determine the terminal pd of the battery.

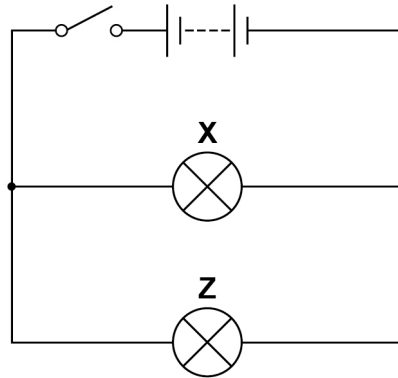
[2 marks]

terminal pd = _____ V



Lamp **Y** fails and is replaced with lamp **Z**, as shown in **Figure 12**.

Figure 12



The switch is closed and both lamps are on.

The resistance of **Z** is greater than $2.40\ \Omega$. The resistance of **X** remains at $2.40\ \Omega$.

The terminal pd is now different from the terminal pd in the circuit in **Figure 11**.

0 8 . 3 Explain this change to the terminal pd.

[2 marks]

0 8 . 4 Explain why the power transferred by **X** in **Figure 12** is different from the power transferred by **X** in **Figure 11**.

[2 marks]

END OF SECTION A

Section B

Answer **all** questions in this section.

*Do not write
outside the
box*

1

1

What is the unit of resistivity in SI fundamental (base) units?

[1 mark]

A $\text{A}^{-2} \text{kg m s}^{-3}$

☐

B $\text{A}^{-2} \text{kg m}^2 \text{s}^{-3}$

☐

C $\text{A}^{-2} \text{kg m}^3 \text{s}^{-3}$

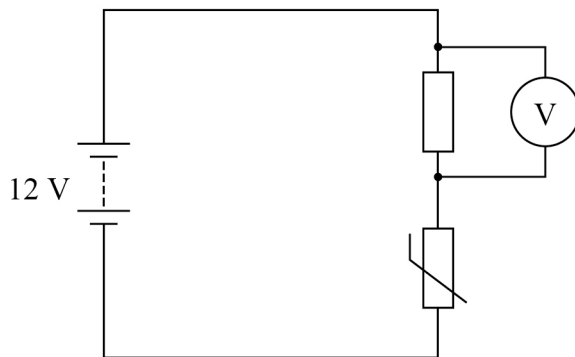
☐

D $\text{A}^2 \text{kg m}^3 \text{s}^{-3}$

☐

Don't write

A circuit contains a battery with an emf of 12 V and negligible internal resistance.



At temperature T_1 , the thermistor has a resistance of $180\ \Omega$ and the voltmeter reading is 3.0 V.

The temperature of the thermistor is changed to T_2 .

The voltmeter reading is now 8.0 V.

What is the resistance of the thermistor at T_2 ?

[1 mark]

A $23\ \Omega$

☐

B $30\ \Omega$

☐

C $90\ \Omega$

☐

D $120\ \Omega$

☐

Turn over for the next question

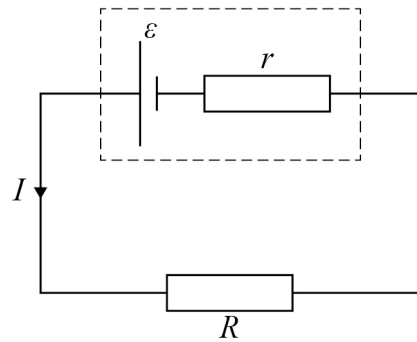
Turn over ►



Do not write

Figure 1 shows a cell in a series circuit with a load resistor. The cell has an internal resistance r .

Figure 1



The emf ε of the cell is given by

$$\varepsilon = IR + Ir$$

where I is the current in the circuit and R is the load resistance.

Explain how this equation is consistent with the law of conservation of energy.

[2 marks]

Turn over for the next question



Don't write

An electrical toy normally needs six cells.

Figure 3a shows the inside of the cell compartment of the toy. The markings show how the cells **A** to **F** should be arranged.

Figure 3b shows the circuit diagram of the cells when they are placed correctly into the compartment.

Figure 3a

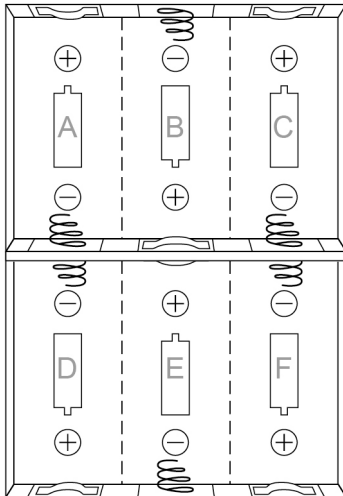
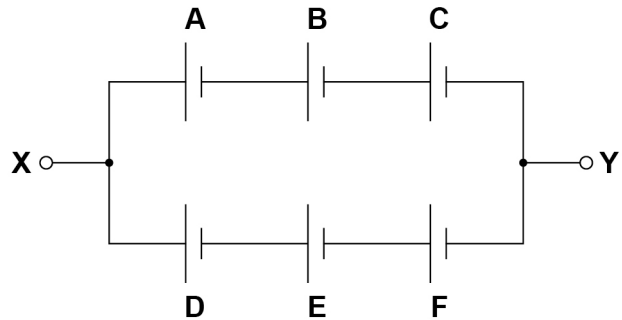


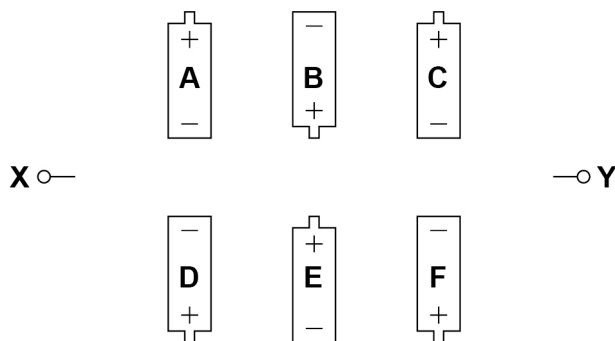
Figure 3b



Draw lines on **Figure 4** to show how the cells should be connected in the compartment.

[1 mark]

Figure 4



Question 5 continues on the next page

Turn over ►



Each cell has an emf of 1.50 V and an internal resistance of $0.25\ \Omega$.

0 5 . 2

On one occasion, a child fits five cells into the toy but leaves out cell **B**.

Suggest and explain the consequences of leaving out cell **B**.

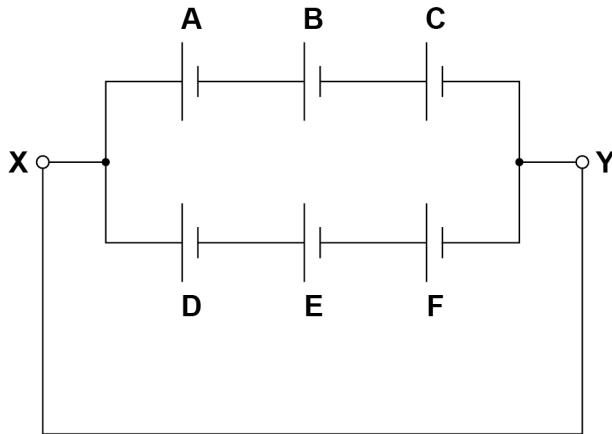
[2 marks]



0 5 . 3

On another occasion, a fault occurs that causes a connection between **X** and **Y** as shown in **Figure 5**. All six cells are fitted correctly. The connection has zero resistance.

Figure 5



Calculate the current in the connection between **X** and **Y**.

[3 marks]

current = _____ A

6

Turn over for the next question

Turn over ►



Do not write

1 2

What is the unit for resistivity in SI fundamental (base) units?

[1 mark]

A $\text{kg m}^3 \text{s}^{-2} \text{A}^{-1}$

☐

B $\text{kg m}^3 \text{s}^{-3} \text{A}^{-1}$

☐

C $\text{kg m}^3 \text{s}^{-2} \text{A}^{-2}$

☐

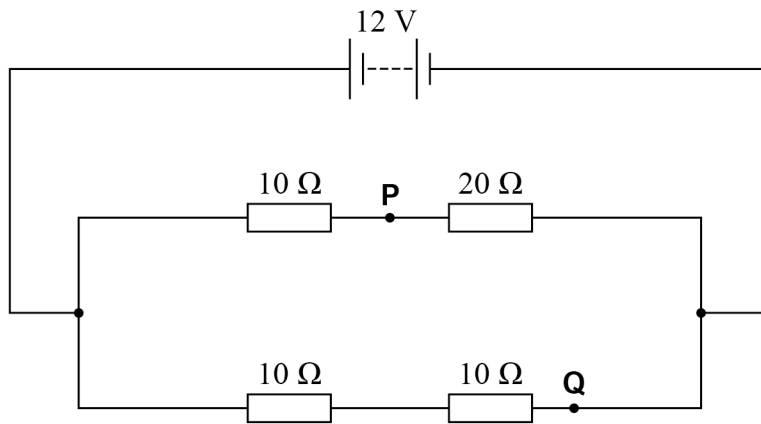
D $\text{kg m}^3 \text{s}^{-3} \text{A}^{-2}$

☐

Don't write

1 3

The circuit shown contains a battery of emf 12 V with negligible internal resistance.



What is the potential difference between **P** and **Q**?

[1 mark]

A 4 V

☐

B 6 V

☐

C 8 V

☐

D 12 V

☐

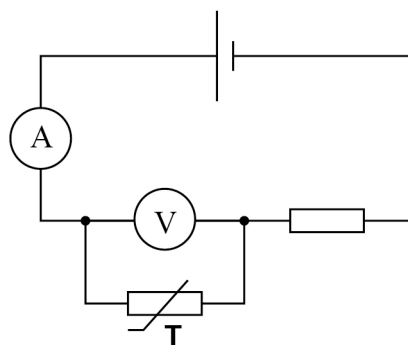
Turn over for the next question

Turn over ►



Do not write

In the circuit shown, the cell has negligible internal resistance.
T is a negative temperature coefficient thermistor.



The temperature of the thermistor changes, causing an increase in the current.

Which row shows the change in temperature and the effect this change has on the voltmeter reading?

[1 mark]

	Change in temperature	Effect on voltmeter reading	
A	increase	increase	☐
B	increase	decrease	☐
C	decrease	increase	☐
D	decrease	decrease	☐

Turn over for the next question



*Do not write
outside the
box*

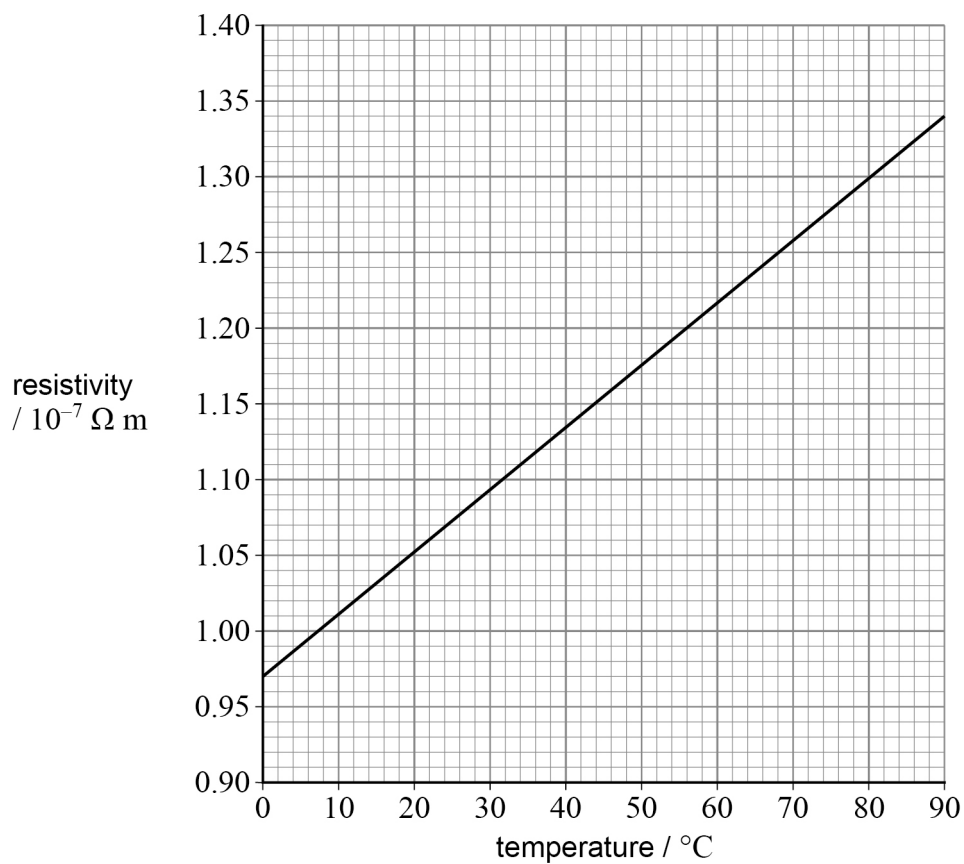
Questions 16 and 17 are about the following experiment.

A student uses a stopwatch to investigate the time period of a pendulum.

There is an absolute uncertainty of 0.2 s at the beginning and the end of each timing.

Figure 5 shows how the resistivity of platinum metal varies linearly with temperature over the range shown.

Figure 5



04.1

A length of platinum wire at a temperature of 0.0°C has a resistance of 99Ω . The diameter of the wire is $1.8 \times 10^{-5} \text{ m}$.

Determine the length of the wire.

[3 marks]

length of wire = _____ m



0 4 . 2

The linear variation of resistivity with temperature shown in **Figure 5** continues up to 200 °C.

Determine the resistivity of platinum when it is at a temperature of 190 °C.

[2 marks]

resistivity = _____ $\Omega \text{ m}$

5

Turn over for the next question

Turn over ►

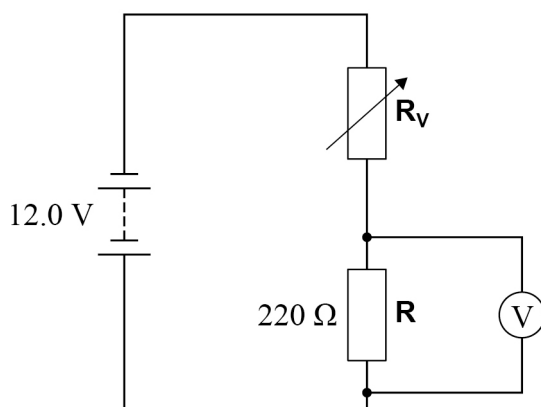


Don't write

0 5

Figure 6 shows a battery connected to a fixed resistor R of resistance $220\ \Omega$ and a variable resistor R_v . The battery has an emf of $12.0\ \text{V}$ and negligible internal resistance.

Figure 6



0 5 . 1

When the resistance of R_v is varied between zero and its maximum, the range of voltmeter readings is $3.0\ \text{V}$ to $12.0\ \text{V}$.

Deduce the maximum resistance of R_v .

[2 marks]

maximum resistance = _____ Ω

0 5 . 2

The battery develops a significant internal resistance.

Explain why a voltmeter reading of $12.0\ \text{V}$ is no longer possible.

[2 marks]

Do not write

09

This question is about different methods of producing white light.

09.1

A fluorescent tube produces white light from ultraviolet (UV) light.

Explain how UV light is produced in a fluorescent tube.

[3 marks]

A type of light-emitting diode (LED) emits UV photons.
White light can be produced from these photons by fluorescence.

09.2

This LED emits UV radiation at a power of 42 mW.
The radiation consists of photons with a frequency of 7.6×10^{14} Hz.

Calculate the number of UV photons emitted each second from the LED.

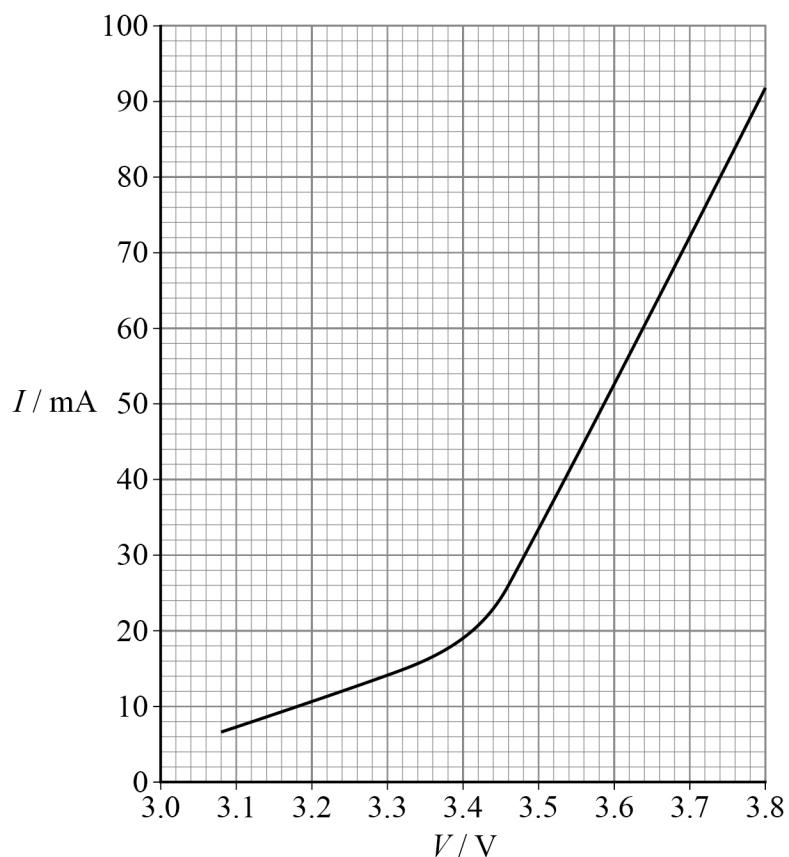
[3 marks]

number of photons emitted each second = _____



Figure 11 shows part of the I - V characteristic graph for the LED.

Figure 11



09.3

The pd across the LED is 3.4 V when it emits UV radiation at a power of 42 mW.

Determine the efficiency of the production of UV radiation in the LED.

[2 marks]

efficiency = _____

Question 9 continues on the next page



09.4

A different source of white light uses a red LED, a green LED and a blue LED.

Figure 12 shows the I - V characteristic graph for each LED.

Figure 12

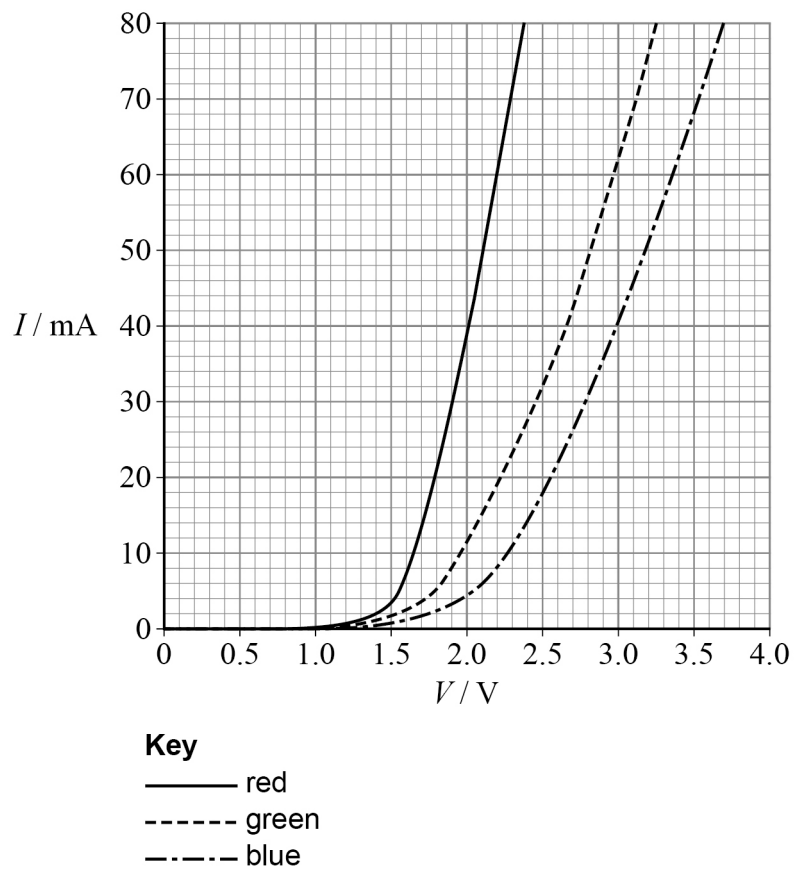
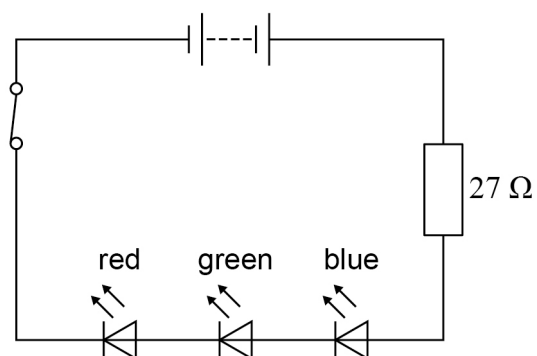


Figure 13 shows the LEDs connected in series to a $27\ \Omega$ resistor and a battery of negligible internal resistance.

Figure 13



The current in the circuit is $43\ \text{mA}$.

Determine the emf of the battery.

[3 marks]

emf = _____ V

END OF SECTION A



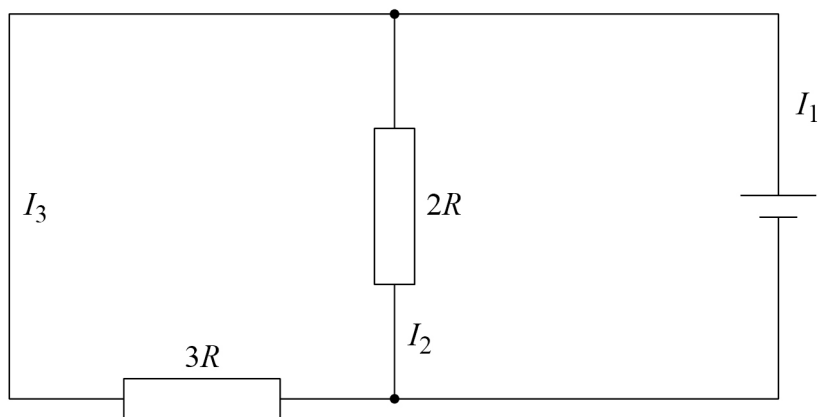
Section B

Answer **all** questions in this section.

*Do not write
outside the
box*

2	4
---	---

Two resistors are connected to a cell that has negligible internal resistance. The resistances and currents are shown on the diagram.



Which relationship, for this circuit, is **not** correct?

[1 mark]

A $3I_1 = 5I_2$ ☐

B $2I_1 = 5I_3$ ☐

C $2I_2 = 3I_3$ ☐

D $5I_1 = 3I_2 + 2I_3$ ☐



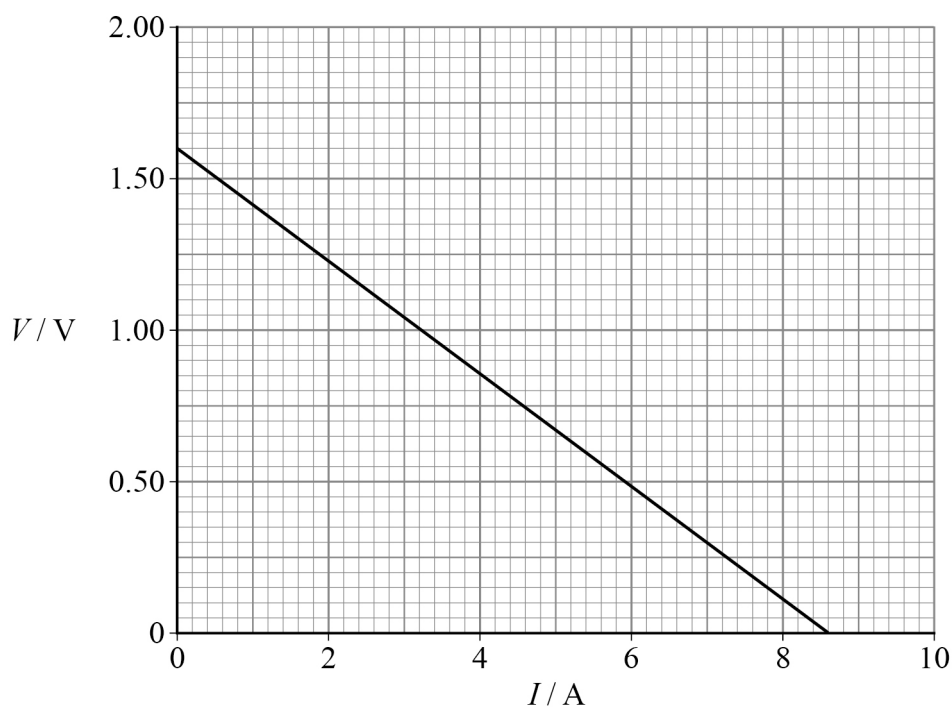
Do not write

0 6

This question is about two electrical cells **A** and **B**.

Figure 7 shows the variation of terminal pd V with current I for cell **A**.

Figure 7

**0 6****1**

Determine the emf of **A**.

[1 mark]

emf = _____ V

0 6**2**

Determine the internal resistance of **A**.

[3 marks]

internal resistance = _____ Ω



0 6 . 3

Cell **B** has the same emf as cell **A**.

B has a much smaller internal resistance than **A**.

A is connected across a lamp.

B is connected across an identical lamp.

Explain any difference between the brightness of the lamps.

Calculations are not required.

[2 marks]

6

Turn over for the next question

Turn over ►



Do not write

A student designs a circuit to indicate the water temperature for a child's bath. A light in the circuit turns on when the bath water is above a certain temperature.

The student puts the circuit inside a floating toy as shown in **Figure 13**.

Figure 13

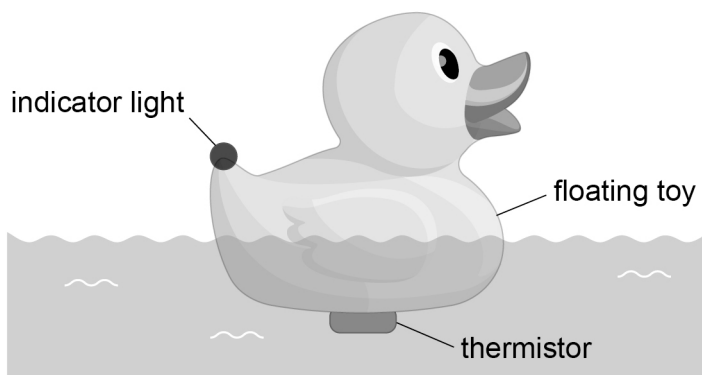
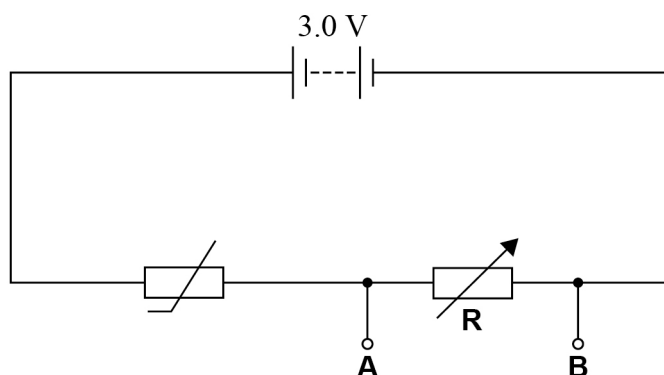


Figure 14 shows part of the circuit. This part contains a battery, a thermistor and a variable resistor **R**.

The battery has an emf of 3.0 V and negligible internal resistance.

Figure 14



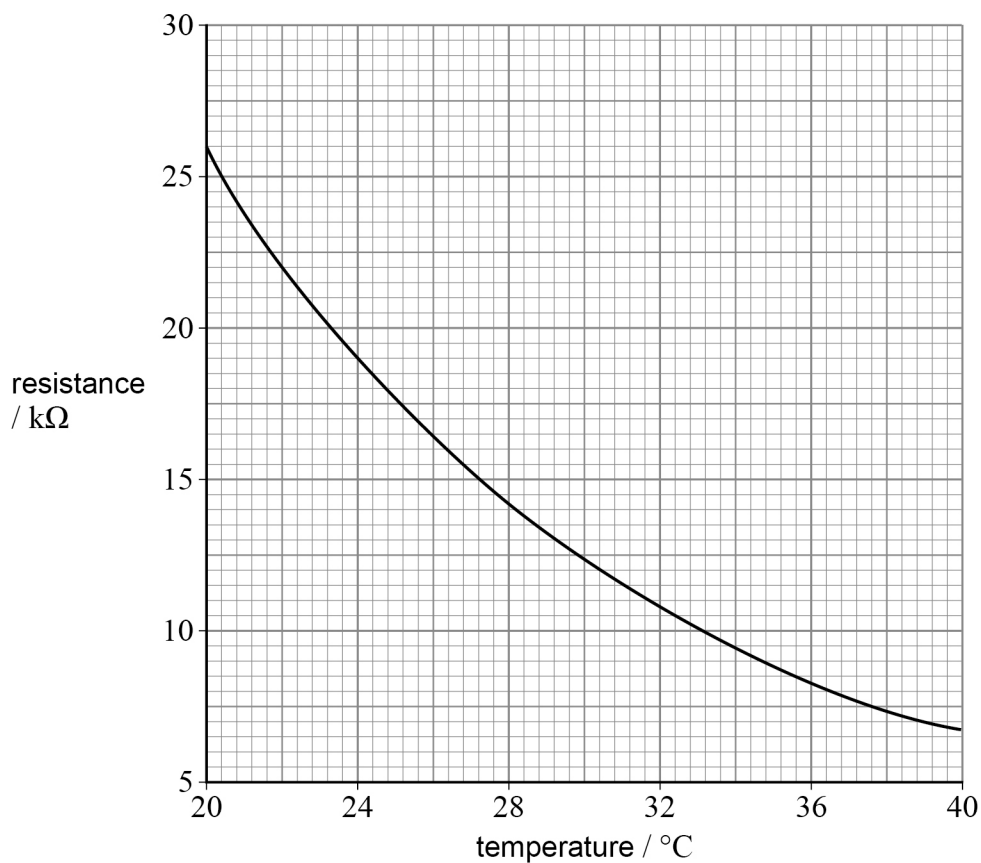
The part of the circuit containing the indicator light is connected between **A** and **B**. This part of the circuit is not shown and does not affect the voltage V_{AB} between **A** and **B**.

Whenever V_{AB} is greater than 0.65 V, the indicator light turns on.



Figure 15 shows the variation of resistance with temperature for the thermistor.

Figure 15



1 1 . 1 The resistance of **R** is set so that V_{AB} is 0.65 V when the temperature is 37 °C.

Calculate the resistance of **R** at this setting.

[3 marks]

resistance = _____ Ω

Question 11 continues on the next page

Turn over ►



1 1 . 2 The student increases the resistance of **R**.

Explain the effect on the temperature at which the indicator light turns on.

[3 marks]

1 1 . 3 Explain the benefit of using **R** in the circuit instead of a resistor of fixed resistance.

[1 mark]

1 1 . 4 Explain whether the part of the circuit between **A** and **B** must have a very high resistance or a very low resistance.

[1 mark]

END OF SECTION B



Section C

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.

For each question, completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown.



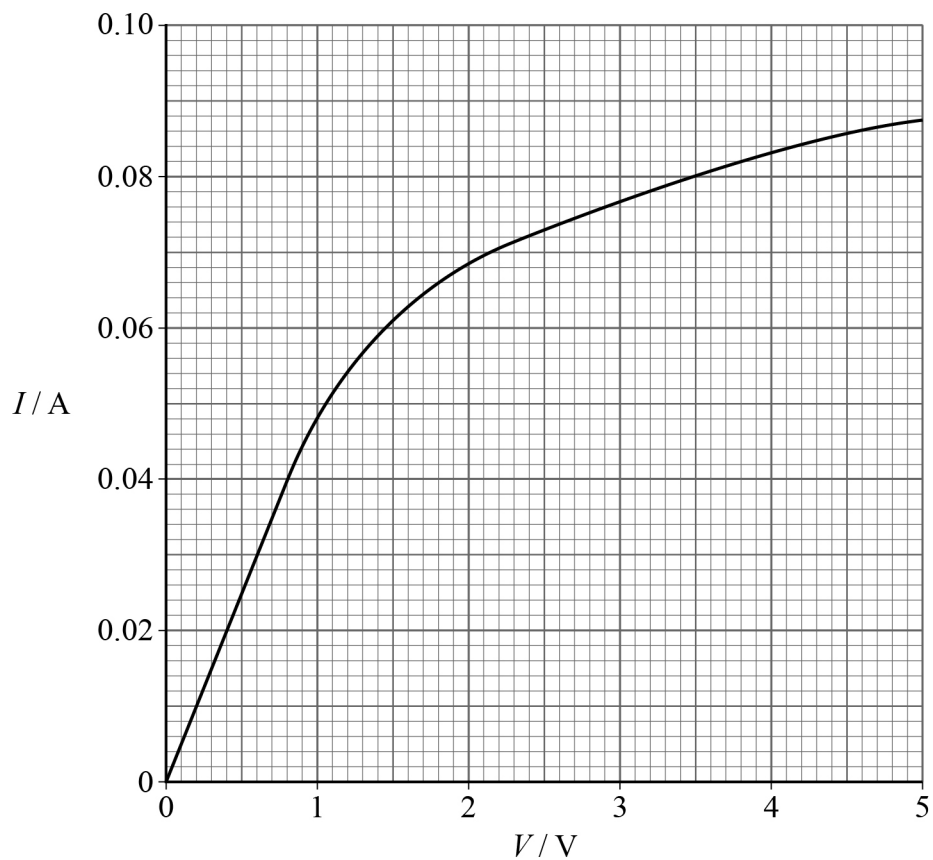
If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown.



You may do your working in the blank space around each question but this will not be marked.
Do **not** use additional pages for this working.

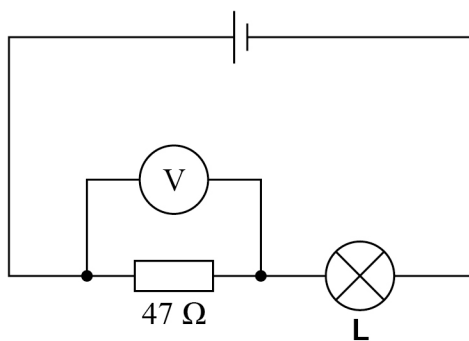
Figure 1 shows the I – V characteristic of a filament lamp **L**.

Figure 1



L is connected in series with a resistor of resistance $47\ \Omega$ and a cell of negligible internal resistance, as shown in **Figure 2**.

Figure 2



0 5 . 1 The voltmeter reading is 2.6 V.

Determine the emf of the cell.

[3 marks]

emf = _____ V

0 5 . 2 A student suggests that **L** shows ohmic behaviour within a voltage range V_1 to V_2 .

Suggest an estimate for V_1 and for V_2 .

Show on **Figure 1** how you obtain your answer.

[2 marks]

V_1 = _____ V

V_2 = _____ V

5



Do not write

0 6

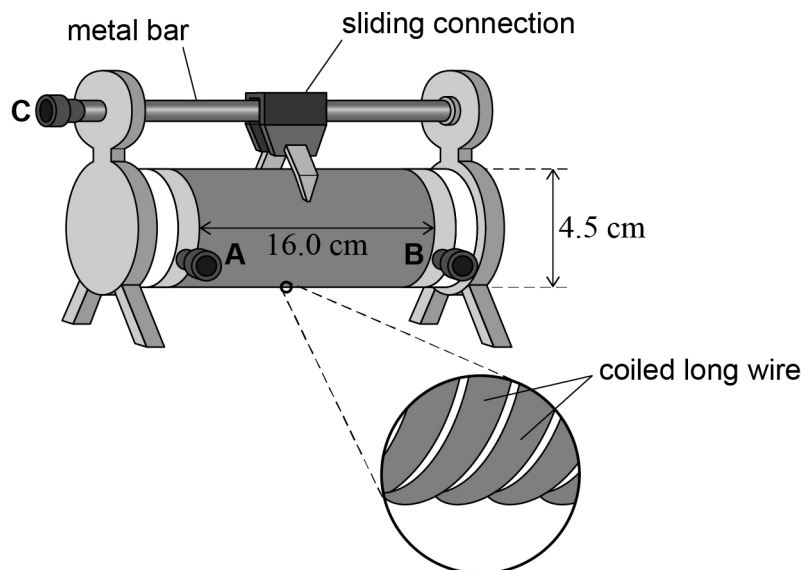
Figure 3 shows a device that can be used as a variable resistor or in a potential divider circuit.

The device has a metal bar and a long thin wire coiled in circular turns around an insulated cylinder.

There are three terminals **A**, **B** and **C**. **A** and **B** are connected to the ends of the wire.

There is a sliding connection that moves along the bar and connects **C** to the wire.

Figure 3



0 6 . 1

The resistance of the metal bar is negligible.

Suggest how the properties of the metal bar and wire ensure that the metal bar has a much smaller resistance than the wire.

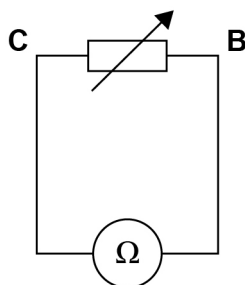
[2 marks]



Terminals **B** and **C** are connected across an ohmmeter as shown in **Figure 4**. In **Figure 4** the device acts as a variable resistor.

When the sliding connection is moved towards **C**, the resistance between **B** and **C** increases.

Figure 4



The maximum resistance measured by the ohmmeter is $18.3\ \Omega$.

0 6 . 2

The coil of wire has a length of 16.0 cm and has 10.0 circular turns per cm . The diameter of each turn is 4.5 cm .

The diameter of the wire is 0.91 mm .

Calculate an estimate for the resistivity of the metal in the wire.

[4 marks]

resistivity = _____ $\Omega\text{ m}$

Question 6 continues on the next page

Turn over ►



0 6 . 3 The resistance of the metal bar is negligible.

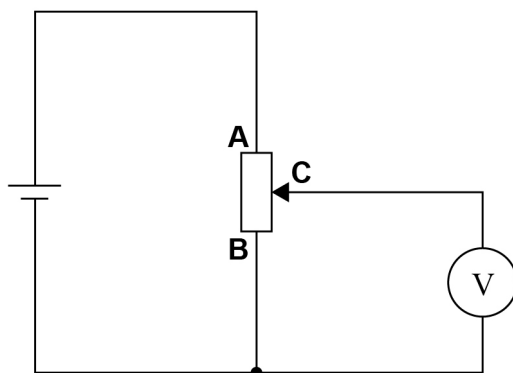
When the sliding connection is at **B**, the ohmmeter accurately measures a non-negligible resistance.

Suggest **one** reason why this resistance measured by the ohmmeter is **not** negligible. **[1 mark]**

0 6 . 4 The device can also be used in a potential divider circuit as shown in **Figure 5**.

Terminals **A**, **B** and **C** are all used in this circuit.

Figure 5



The emf of the cell is 1.6 V and the internal resistance of the cell is 0.95 Ω .

Calculate the maximum possible reading on the voltmeter in this circuit.

Assume that the maximum resistance of the device is 18.3 Ω .

[2 marks]

maximum reading = _____ V

Turn over for the next question

*Do not write
outside the
box*

**DO NOT WRITE ON THIS PAGE
ANSWER IN THE SPACES PROVIDED**

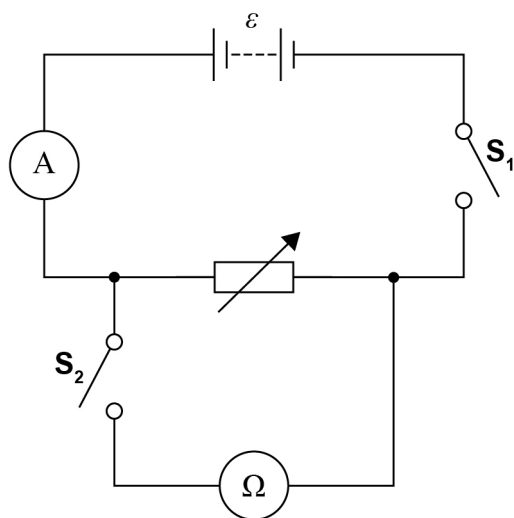
Turn over ►



Don't write

A student uses the circuit shown in **Figure 11** to determine the emf ε and the internal resistance r of a battery.

Figure 11



A variable resistor is used as the load resistance R in the circuit. It is adjusted to obtain different values of R .

With S_1 open and S_2 closed, the value of R is set and measured using the ohmmeter. S_2 is now opened and then S_1 is closed, and the current I is recorded. This is repeated for different values of R .

Table 1 shows some of the student's data.

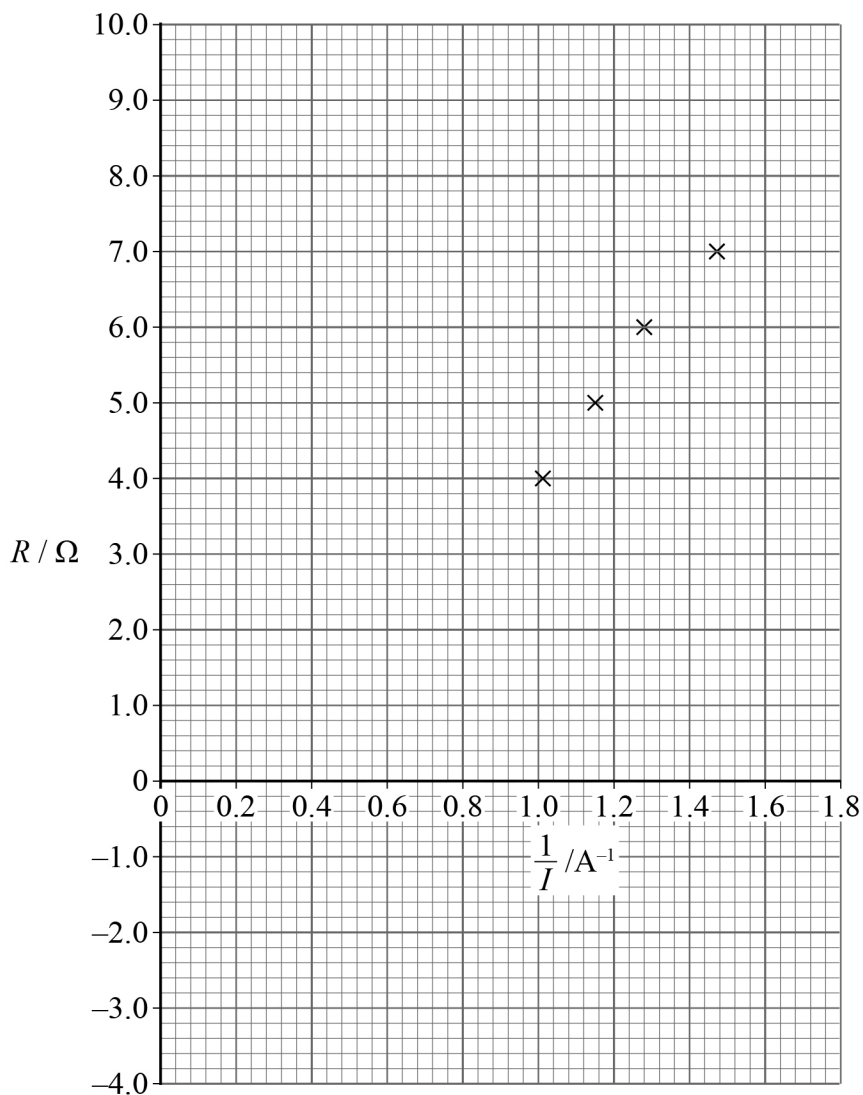
Table 1

R / Ω	I / A	$\frac{1}{I} / \text{A}^{-1}$
4.0	0.991	1.01
5.0	0.872	1.15
6.0	0.781	1.28
7.0	0.680	1.47
8.0	0.611	
9.0	0.560	



Figure 12 shows the first four points of a graph of R against $\frac{1}{I}$.

Figure 12



Complete **Table 1**.

Go on to plot the last two points on **Figure 12**.

[2 marks]

Draw on **Figure 12** a line of best fit.

You should extend your line to the y -axis.

[1 mark]

Question 10 continues on the next page

Turn over ►

1 0 . 3

Show algebraically that:

- gradient = ε
- y-intercept = $-r$.

[2 marks]

1 0 . 4

Determine ε and r .

[3 marks]

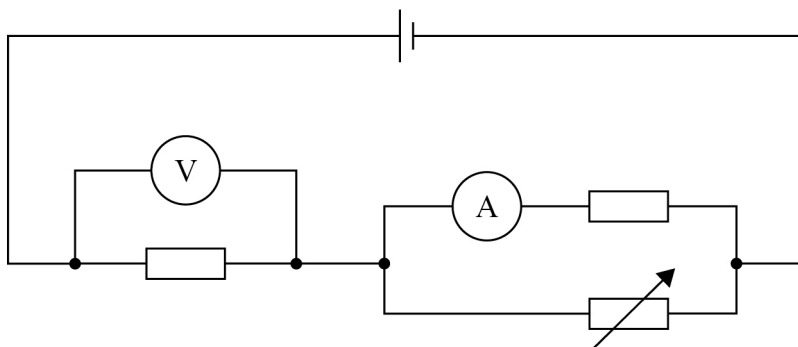
$$\begin{aligned}\varepsilon &= \text{_____ V} \\ r &= \text{_____ } \Omega\end{aligned}$$

8



Do not write

The circuit shown contains two fixed resistors, a variable resistor, and a cell with negligible internal resistance.



The resistance of the variable resistor is increased.

Which row shows what happens to the voltmeter reading and the ammeter reading?

[1 mark]

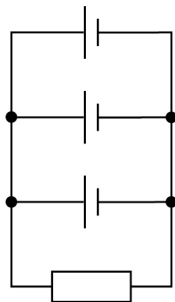
	Voltmeter reading	Ammeter reading	
A	increases	stays the same	☐
B	increases	increases	☐
C	decreases	stays the same	☐
D	decreases	increases	☐

Turn over for the next question



Do not write

A resistor of resistance R and three identical cells of emf ε and internal resistance r are connected together as shown.



What is the potential difference across the resistor?

[1 mark]

A $\frac{\varepsilon R}{R + 3r}$

B $\frac{\varepsilon R}{3R + r}$

C $\frac{3\varepsilon R}{R + 3r}$

D $\frac{3\varepsilon R}{3R + r}$